REFERENCE GUIDE

Standard Probability

Foundations: Counting, Conditional Probability, and Bayes' Theorem

This guide covers the foundational core of the standard probability curriculum used in the BBE exam format, written for someone meeting each idea for the first time: sample spaces and counting, the complement rule, unions and inclusion-exclusion, conditional probability, independence, sampling with and without replacement, combinatorial probability, the law of total probability with Bayes' theorem, and discrete probability distributions. Each section builds up the theory gradually in plain language, starting from a simple everyday idea before introducing any formula, explains how to recognize that type of question on sight, shows every required formula rendered clearly, includes a diagram wherever the topic is naturally visual, and works through at least one fully solved numerical example. A step-by-step method is given for approaching exam questions on each subtopic, and a complete formula cheat sheet is included at the end.

BBE School — Mathematics

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01 Sample Spaces and Classical Probability

How to recognize this type of question

The scenario involves a fair, unweighted process — a fair die, fair coin, well-shuffled deck, or randomly picked item — and simply asks for the probability of a single, direct event, with no conditioning language ("given that"), no "at least," and no mention of removing items without replacement. If the question can be answered purely by counting favorable outcomes over total outcomes, start here.

Probability is just a way of measuring how likely something is to happen, using a number between 0 and 1 (or, written as a percentage, between 0% and 100%). A probability of 0 means the thing can never happen; a probability of 1 (100%) means it is certain to happen; a probability of 0.5 (50%) means it is exactly as likely to happen as not.

Every probability question starts from a random experiment — an action with an uncertain result, such as rolling a die, flipping a coin, or drawing a card. The sample space, written S, is simply the list of every single thing that could possibly happen. For one coin flip, the sample space is {Heads, Tails}. For one roll of a six-sided die, the sample space is {1, 2, 3, 4, 5, 6}. An event is any collection of outcomes we happen to be interested in — for example, "rolling an even number" is the event {2, 4, 6}, made up of three of the six outcomes in the sample space.

When every outcome in the sample space is equally likely to occur — a fair coin, a fair die, a well-shuffled deck — probability turns into nothing more than counting. Ask yourself two questions: "how many outcomes am I interested in?" and "how many outcomes are there in total?" The answer to the first question, divided by the answer to the second, is the probability. This is called classical probability, and it is the starting point for almost everything else in this guide.

CLASSICAL PROBABILITY (EQUALLY LIKELY OUTCOMES)

$$P(E) = \frac{n(E)}{n(S)}$$

In plain words: P(E) means "the probability that event E happens." n(E) is how many outcomes count as a success for event E. n(S) is the total number of possible outcomes. This formula only works when every individual outcome is equally likely — it cannot be used directly if some outcomes are more likely than others (a weighted die, for instance).

Before moving to a harder example, it helps to see the simplest possible case. A single fair die has 6 equally likely outcomes: 1, 2, 3, 4, 5, 6. Suppose we want P(rolling a 4 or higher). The outcomes that count are {4, 5, 6} — that is 3 outcomes out of 6 total, so the probability is 3/6 = 0.5, or 50%. Once a sample space has more than one item changing at a time — such as two dice rolled together — the same idea still applies, but the counting takes a little more care, which is where a grid like the one below becomes useful.

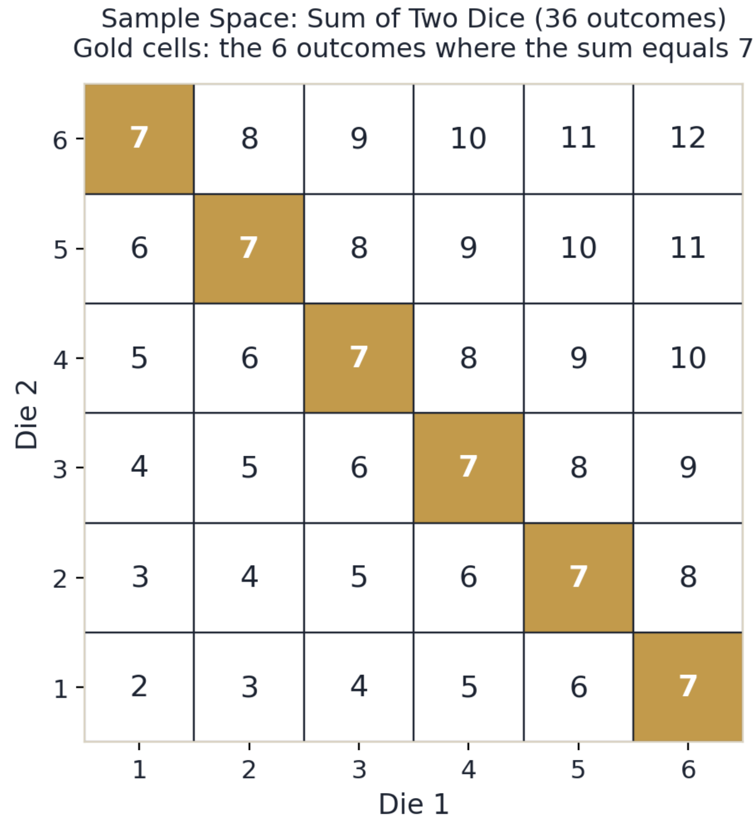

Figure 1.1 — All 36 equally likely outcomes for two dice. Each cell shows the sum for that pair; the 6 gold cells are the outcomes where the sum equals 7.

How to solve this type of question

1. Work out what counts as a single outcome (for two dice, a single outcome is a pair, such as "die 1 shows 3 and die 2 shows 5"), and count the total number of outcomes in the sample space, n(S).
2. Identify precisely which outcomes count as the event in question, and count them: n(E). For a small sample space, it genuinely helps to list every outcome out by hand, or use a grid like Figure 1.1, rather than trying to count "in your head."
3. Divide n(E) by n(S) to get the probability, as a decimal.
4. Convert the decimal to a percentage (multiply by 100) and compare it against whatever threshold the exam statement gives.

EXAMPLE 1.1

Two fair six-sided dice are rolled once. There are 36 equally likely ordered outcomes (shown in Figure 1.1, since each of the 6 faces of the first die can pair with each of the 6 faces of the second die, giving 6×6 = 36 combinations). Find P(sum = 7).

Ordered pairs summing to 7: (1,6)(2,5)(3,4)(4,3)(5,2)(6,1)	**6 outcomes**
Total outcomes	**36**
P(sum = 7) = 6 / 36	**0.1667**

Result: P(sum = 7) ≈ 16.67%

Counting shortcut for two dice

For two fair dice, the number of ordered pairs summing to k is $(k - 1)$ for $2 \le k \le 7$, and $(13 - k)$ for $7 < k \le 12$. In Figure 1.1, this is just counting how many cells lie along each diagonal line running from bottom-left to top-right — it lets you find any sum-count without listing every pair by hand. For example, for a sum of 4, the shortcut gives $(4-1) = 3$, matching the three pairs (1,3), (2,2), (3,1).

02 The Complement Rule

How to recognize this type of question

The exam wording contains the phrase "at least one," "at least once," "never," or asks for the probability that something fails to happen across several repeated attempts. Any time computing the direct question would require adding up many separate cases (exactly one, exactly two, exactly three…), it is a strong sign the complement rule is the intended shortcut.

The complement of an event A, written A^c (read "A complement" or "not A"), simply means "A does not happen." Think of any event as splitting the whole sample space into two non-overlapping piles: the outcomes where A happens, and the outcomes where A does not happen. Every single outcome falls into exactly one of these two piles — never both, never neither. Because the two piles together make up 100% of the possibilities, their probabilities must add up to exactly 1. This is true for every event, with no exceptions and no conditions to check first.

This one simple fact turns out to be an extremely useful shortcut for a very common type of exam question: "what is the probability that at least one of several things happens?" Computing "at least one" directly is often painful, because it usually means adding up the probability of exactly one happening, plus exactly two, plus exactly three, and so on — many separate cases to work out and add together. But notice that the complement of "at least one" is simply "none at all" (zero of them) — and "none at all" is almost always a single, easy calculation. So instead of solving the hard question directly, we solve the easy opposite question, and then subtract from 1.

THE COMPLEMENT RULE AND ITS MOST COMMON USE

$$P(A^c) = 1 - P(A)$$

In plain words: the probability that A does not happen is 1 minus the probability that A does happen. If you know one, you immediately know the other.

"AT LEAST ONE" VIA THE COMPLEMENT

$$P(\text{at least one}) = 1 - P(\text{none})$$

The second line is just the first rule applied to a repeated experiment: "at least one success across several trials" is the complement of "zero successes across all of them." So P(zero successes) is the one thing you actually need to calculate.

Why is P(zero successes) usually easy to find? Because "zero successes across several repeated, independent trials" means every single trial failed, one after another. When trials are independent (covered fully in Section 5), the probability that they all happen is found by simply multiplying each trial's own probability together — which is exactly what the exponent in the formula above is doing.

How to solve this type of question

1. Read the statement carefully and identify the single opposite outcome. If the statement says "at least one X," the opposite is "no X at all" (zero of them) — write this opposite outcome down explicitly before doing any arithmetic.
2. Compute the probability of that opposite outcome directly. For repeated trials, this usually means multiplying the "failure" probability by itself once for each trial.
3. Subtract the result from 1.
4. Compare the answer to the threshold given in the exam statement.

EXAMPLE 2.1

A fair coin is flipped 5 times. Find P(at least one tail among the 5 flips).

The opposite of "at least one tail" is "all 5 flips are heads"	–
P(all heads) = $(1/2)^5 = (1/2)\times(1/2)\times(1/2)\times(1/2)\times(1/2)$	**0.03125**
P(at least one tail) = 1 − 0.03125	**0.96875**

Result: P(at least one tail) ≈ 96.88%

Default to the complement

Any exam statement containing the phrase "at least one" should immediately make you think of this rule. It is very rarely faster to compute "at least one" any other way, so treat the complement as your automatic first move for this kind of wording, rather than a special trick you only remember occasionally.

03 Unions and the Inclusion-Exclusion Principle

How to recognize this type of question

The question mentions two or more overlapping categories or groups (people using multiple apps, students taking multiple subjects, components with multiple properties) and asks for "at least one," "or," "exactly one," or "exactly two" among them. A Venn diagram, or a list of individual, pairwise, and triple-overlap counts, is a strong giveaway that this section applies.

A Venn diagram is simply a picture used to show how groups of things overlap: each event is drawn as a circle, and anything that belongs to more than one event at the same time is shown by drawing the circles so they overlap in the middle. This section uses Venn diagrams because they make it very easy to see the double-counting problem that this section solves.

Suppose you want the probability that at least one of two events, A or B, happens — in Venn-diagram terms, the probability of landing anywhere inside either circle. A natural first guess is to add P(A) + P(B). But if some outcomes belong to both A and B — the overlapping middle region — that overlap has now been counted twice: once inside P(A), and again inside P(B). The inclusion-exclusion principle fixes this by subtracting the overlap once, so every outcome ends up counted exactly one time, no more and no less.

A quick, small example makes this concrete before moving to three events. Suppose P(A) = 0.5, P(B) = 0.4, and P(A and B) = 0.2. Adding P(A) + P(B) gives 0.9, but this counts the overlapping 0.2 twice. Subtracting it once gives the correct answer: P(A or B) = 0.5 + 0.4 − 0.2 = 0.7.

With three events, the same double-counting problem shows up again, but needs one extra correction. Adding P(A) + P(B) + P(C) counts every pairwise overlap (A-and-B, A-and-C, B-and-C) twice, so all three pairwise overlaps must be subtracted once each. But now the very middle region — outcomes in all three events at once — was counted three times in the original sum (once in each of P(A), P(B), P(C)) and then subtracted three times by the three pairwise corrections, leaving it removed completely. So it has to be added back once, to bring it back to being counted exactly one time overall, just like every other region.

UNION OF TWO EVENTS

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

In plain words: to find the chance that at least one of two events happens, add their individual probabilities and then subtract the overlap once, so it is not double-counted.

UNION OF THREE EVENTS

$$P(A \cup B \cup C) = P(A)+P(B)+P(C) - P(A\cap B) - P(A\cap C) - P(B\cap C) + P(A\cap B\cap C)$$

In plain words: add the three individual probabilities, subtract each of the three pairwise overlaps once, then add the very middle (all-three) overlap back once.

EXACTLY ONE OF THREE EVENTS ("ONLY X")

$$\text{Only X} = |X| - |X\cap Y| - |X\cap Z| + |X\cap Y\cap Z|$$

This finds the region belonging to X alone, with no overlap with Y or Z at all.

EXACTLY TWO OF THREE EVENTS, NOT ALL THREE

$$\text{Pairwise only}(X, Y) = |X \cap Y| - |X \cap Y \cap Z|$$

Sum the three "only-X" values across all three events to get P(exactly one); sum the three pairwise-only values to get P(exactly two, but not all three).

Three-Set Venn Diagram (Example 3.1)

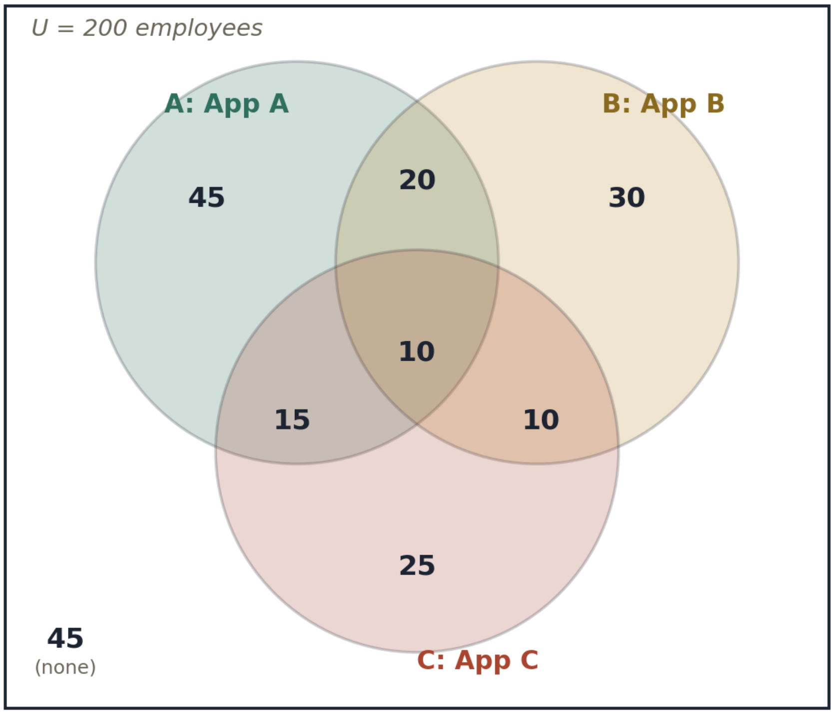

Figure 3.1 — A three-set Venn diagram for Example 3.1. Each region shows how many of the 200 employees fall into that exact combination of App A, B, and C usage.

How to solve this type of question

1. List every given value: the size (or probability) of each individual set, each pairwise overlap, and the triple overlap.
2. For "at least one," apply the union formula directly.
3. For "exactly one" or "exactly two, not three," build the "only" regions first using the formulas above — do not just guess a region's size from the diagram without applying the formula, since raw overlap values usually include the triple-overlap region and need to have it subtracted out.
4. Divide the resulting count by the total population size to convert it into a probability.

EXAMPLE 3.1 — worked calculation

Among 200 employees: 90 use App A, 70 use App B, 60 use App C. 30 use both A and B, 25 use both A and C, 20 use both B and C, and 10 use all three (see Figure 3.1). Find P(at least one app), P(exactly one), and P(exactly two, not three).

$\lvert A \cup B \cup C \rvert = 90+70+60-30-25-20+10$	**155**
P(at least one) = 155 / 200	**77.5%**
Only A = 90−30−25+10; Only B = 70−30−20+10; Only C = 60−25−20+10	**45, 30, 25**
P(exactly one) = (45+30+25) / 200	**50.0%**
$A \cap B$ only = 30−10; $A \cap C$ only = 25−10; $B \cap C$ only = 20−10	**20, 15, 10**
P(exactly two, not three) = (20+15+10) / 200	**22.5%**

Region values on a diagram are already "only" values

When a Venn diagram is drawn with a number written inside each distinct region (as in Figure 3.1), those numbers are typically already the "only" values — the triple-overlap region is its own separate number, not included in the pairwise regions. The only-X formula matters most when you are instead given raw overlap totals $|A \cap B|$ that still include the triple overlap, and need to subtract it out yourself.

04 Conditional Probability

How to recognize this type of question

The wording explicitly restricts the situation with a phrase such as "given that," "if," "knowing that," or shows the vertical-bar notation $P(A \mid B)$. The question is not asking about the full original sample space anymore — it has already told you something happened, and wants a probability calculated only among the cases where that is true.

Conditional probability answers a "given that" question: if we already know event B happened, how likely is it that event A also happened, or will happen? Learning that B occurred effectively shrinks the sample space we are working with — instead of considering every single outcome in S, we now only consider the outcomes inside B, and ask what fraction of those also happen to lie inside A.

A helpful everyday analogy: "what is the probability it rains today" is a different question from "what is the probability it rains today, given that it is currently cloudy." The second question throws away every day that was not cloudy, and only looks at the cloudy days, asking what fraction of those cloudy days turned out rainy. That fraction is usually noticeably larger than the plain, unconditional chance of rain, because clouds are linked to rain — this is exactly what conditioning captures mathematically.

A very small numerical example: suppose in a group of 20 people, 8 people own a dog, and of those 8 dog owners, 5 also own a cat. If someone is picked at random from the dog owners specifically, P(cat owner | dog owner) = 5/8 = 0.625. Notice that the denominator, 8, is the number of dog owners — not the full group of 20 — because "given that they own a dog" has already restricted our attention to that smaller group.

CONDITIONAL PROBABILITY

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

In plain words: to find P(A given B), take the probability of both A and B happening together, and divide by the probability of B alone. B is the "given" condition, so its probability always ends up on the bottom.

CONDITIONAL PROBABILITY FOR EQUALLY LIKELY OUTCOMES

$$P(A \mid B) = \frac{|A \cap B|}{|B|}$$

P(B) (or the count of B) must be nonzero for the conditional probability to be defined — you cannot condition on something impossible. Note carefully: P(A | B) is generally not the same number as P(B | A) — the order in which the two events are written matters a great deal.

How to solve this type of question

1. Identify the conditioning event B — the thing that is "given," written after the vertical bar — and find its count or probability. This becomes your new denominator.
2. Identify the overlap between A and B: the outcomes that satisfy both conditions at the same time.
3. Divide the overlap by P(B) (or by the count of B, when every outcome is equally likely).
4. Sanity check the result: it should represent a fraction of B specifically, not a fraction of the entire original sample space. If your answer is larger than 1, or you divided by the wrong total, you have mixed this up.

EXAMPLE 4.1 — worked calculation

Two fair dice are rolled. Given that the product of the two dice is odd, find P(the two dice show the same number).

Outcomes with odd product (both dice must individually be odd): 3 × 3	**9 outcomes**
Of these, doubles: (1,1), (3,3), (5,5)	**3 outcomes**
P(same number | odd product) = 3 / 9	**0.3333**

Result: P(same number | odd product) = 1/3 exactly

P(A|B) is not the same as P(B|A)

Conditioning on "the second card is a Queen" and then asking about the first card is a different calculation from conditioning on the first card and asking about the second. Without replacement, removing a specific card changes the size of the remaining pool, and this can shift a related probability in either direction. Always recompute the conditional directly from its definition rather than assuming the two directions must give the same answer.

05 Independent Events

How to recognize this type of question

The problem explicitly states that two events are "independent," or describes two physically separate, unrelated processes (two different machines, two unrelated coin flips, two different customers), and asks for the probability of both happening, or provides a joint-probability table and asks you to check whether the variables are independent.

Two events are independent if knowing that one of them occurred gives you no new information whatsoever about the other. Flipping a coin twice is the classic example: whatever happened on the first flip — heads or tails — the second flip still has exactly a 50/50 chance of either outcome. The first flip does not "remember" anything, and it does not influence the coin in any way.

It is important not to confuse independence with mutual exclusivity, which is a completely different (and, in a sense, opposite) idea. Mutually exclusive events cannot both happen at the same time — so if you learn that one of them occurred, you know for certain that the other one did not. That is about as far from "no new information" as it is possible to get: learning one happened tells you everything about the other (namely, that it definitely did not happen). Because of this, two mutually exclusive events that both have a nonzero chance of occurring can never be independent of each other — the two properties are, in this sense, at odds with one another.

THE INDEPENDENCE CONDITION

$$P(A \cap B) = P(A) \times P(B)$$

In plain words: if two events are independent, the chance of both happening together is just the product of their two individual chances — nothing more complicated than simple multiplication.

AN EQUIVALENT WAY TO STATE INDEPENDENCE

$$P(A \mid B) = P(A)$$

This says the same thing in a different way: if A and B are independent, then conditioning on B does not change the probability of A at all — it stays exactly as it was before you knew anything about B. To test whether two events (or two variables in a table) are independent, compare P(A ∩ B) to P(A) × P(B): if the two numbers match, the events are independent; if even one comparison fails to match, they are not.

How to solve this type of question

1. Determine whether the problem tells you the events are independent, describes physically separate/unrelated processes (which usually implies independence), or gives you a table of joint probabilities to check.
2. If independence is confirmed, "and" probabilities become simple products: P(A and B) = P(A) × P(B).
3. For "or" questions between independent events, still use the general union rule from Section 3 — independence does not change how unions work, only how the intersection term is computed.
4. For "neither" questions, either subtract the union from 1, or compute directly as (1−P(A))×(1−P(B)) — both routes should agree, giving you a useful cross-check.

EXAMPLE 5.1 — worked calculation

Events A and B are independent, with P(A) = 0.35 and P(B) = 0.60. Find P(A and B), P(A or B), P(neither), and P(exactly one).

P(A ∩ B) = 0.35 × 0.60	**0.21**
P(A ∪ B) = 0.35 + 0.60 − 0.21	**0.74**
P(neither) = 1 − 0.74 (cross-check: 0.65 × 0.40 = 0.26, matches)	**0.26**
P(exactly one) = P(A∪B) − P(A∩B) = 0.74 − 0.21	**0.53**

Independence means conditioning changes nothing

For independent events, P(A | B) always equals the plain, unconditional P(A) — learning B occurred does not raise or lower the probability of A. Any exam statement claiming that a conditional probability is strictly greater than (or less than) the corresponding unconditional probability is automatically false whenever the two events are independent, with no further calculation needed to check it.

06 Probability Without Replacement

How to recognize this type of question

Items are drawn or dealt one after another (cards, marbles, raffle tickets, people) and the wording says or implies "without replacement," "not returned," or simply describes a sequence of draws from a shrinking pool with nothing put back. If a second or third draw is being discussed from the same, ever-smaller group, this is the section to use.

Many problems draw several items one after another from a group, without putting each item back before the next draw — dealing cards from a deck, drawing marbles from a bag, picking raffle tickets from a box. Because each item removed is gone for good, the pool available for the next draw is smaller than it was before. And if the first item removed happened to match the category we care about, the count of matching items left in the pool shrinks too, not just the total.

This is the key difference from Section 5: draws without replacement are not independent. What happens on the first draw genuinely changes the odds for the second draw, because the pool itself has physically changed.

A simple example to build intuition: a bag contains 4 red marbles and 6 blue marbles (10 total). P(first marble is red) = 4/10. If that first marble is not put back, only 9 marbles remain, and only 3 of them are red (since one red marble was just removed) — so P(second marble is also red, given the first was red) = 3/9. To find P(both marbles are red), multiply the two fractions together: (4/10) × (3/9).

TWO SEQUENTIAL DRAWS WITHOUT REPLACEMENT

$$P(\text{both from X}) = \frac{\text{count}}{\text{total}} \times \frac{\text{count} - 1}{\text{total} - 1}$$

In plain words: the first fraction is the ordinary probability for the first draw. The second fraction reflects that one matching item, and one item overall, have both already been removed from the pool.

ONE OF EACH OF TWO SPECIFIC CATEGORIES, IN EITHER ORDER

$$P(\text{one X, one Y}) = 2 \times \frac{X}{\text{total}} \times \frac{Y}{\text{total} - 1}$$

On the second draw, both the numerator (count remaining) and the denominator (total remaining) have shrunk by exactly 1 compared to the first draw. The factor of 2 out front accounts for the fact that category X could come first and Y second, or Y could come first and X second — both orders satisfy "one of each."

How to solve this type of question

1. Write the probability of the first draw as a simple fraction: count of interest over total population.
2. For the second draw, reduce both the numerator and the denominator by however many items were removed on the first draw (usually by 1 for a single item).
3. Multiply the fractions across all draws in the sequence.
4. If the order of the two categories does not matter ("one of each, either order"), multiply your answer by 2 (or, more generally, by the number of valid orderings).

EXAMPLE 6.1 — worked calculation

Two cards are drawn without replacement from a standard 52-card deck. Find P(both cards are hearts) and P(both cards are the same color).

P(both hearts) = (13/52) × (12/51)	**0.05882**
P(both same color) = 2 × (26/52) × (25/51)	**0.49020**

Result: P(both hearts) ≈ 5.88%; P(same color) ≈ 49.02%

The same-color shortcut

When two categories (such as the two card colors) are of equal size, P(same category) is simply double P(both a specific one) — there is no need to compute "both red" and "both black" separately and add them, since by symmetry they are identical.

07 Combinatorial Probability

How to recognize this type of question

A whole sample or group is chosen all at once, with no order involved — a "committee," "team," "hand of cards," or "sample" selected from a larger population that is split into subgroups (e.g., women/men, red/blue). Look for the words "selected," "chosen," or "randomly formed" applied to a group, rather than a sequence of individual draws described one at a time.

Section 6 handled items drawn one at a time, in a specific sequence. This section handles the same underlying idea, but for a sample of several items chosen all at once, as a single group, with no order attached — a committee selected from a larger group, or a hand of cards dealt together all in one go. The tool for counting "how many ways" here is called a combination, written C(n, r), meaning the number of different ways to choose r items out of n total items, where the order they are chosen in does not matter at all.

It is easiest to understand C(n, r) by building it up in two small steps, rather than memorizing it as a single block.

Step 1: count the orderings where sequence does matter

Suppose order did matter for a moment — imagine lining up r chosen items in a specific order, like awarding 1st place, 2nd place, and so on. For the first position, there are n items to choose from. For the second position, only (n−1) remain, since one has already been used. This continues until r positions are filled. The result is called a permutation, P(n, r):

PERMUTATIONS — ORDER MATTERS

$$P(n, r) = \frac{n!}{(n-r)!}$$

n! ("n factorial") means n × (n−1) × (n−2) × … × 1. Dividing by (n−r)! simply stops the multiplication after r terms, leaving only n × (n−1) × … down to r terms.

Step 2: remove the duplicate orderings, since order does not actually matter

A combination does not care about order — choosing items {A, B, C} for a committee is the same result whether A, B, or C was picked first. But the permutation count from Step 1 treated every different ordering of the same r items as a separate outcome. Each group of r items was counted once for every possible order those same r items could be arranged in, which is r! (r factorial) times. So, to go from "order matters" to "order does not matter," the permutation count is simply divided by r!, removing exactly that duplication.

C(N, R) — THE COMBINATION FORMULA ("N CHOOSE R")

$$C(n, r) = \frac{n!}{(n-r)!\, r!}$$

In plain words: first count every ordered arrangement of r items out of n (Step 1), then divide by r! to collapse every group of duplicate orderings back down to a single count. The result, C(n, r), is exactly how many ways there are to choose r items from n when the order they come in does not matter.

A small worked check confirms this: C(4, 2), the number of ways to choose 2 items from 4 (call them 1, 2, 3, 4), should be 6. Using the two-step build-up: P(4,2) = 4 × 3 = 12 ordered pairs (12, 21, 13, 31, 14, 41, 23, 32, 24, 42, 34, 43). Dividing by 2! = 2 (since each unordered pair, like {1,2}, was counted twice — once as "12" and once as "21") gives 12 / 2 = 6, matching the 6 distinct unordered pairs: {1,2}, {1,3}, {1,4}, {2,3}, {2,4}, {3,4}. In practice, for exam purposes, C(n,r) values are usually computed directly with a calculator rather than by hand-expanding every factorial — the derivation above is what to understand, not what to repeat by hand every time.

The formulas below apply this same C(n, r) tool to a population split into two subgroups (for example, choosing a committee from a mix of women and men), where the notation switches to k for "how many are drawn from the first subgroup" simply to keep track of which count is which — it is still exactly the same combination operation.

CHOOSING A MIXED SAMPLE FROM TWO SUBGROUPS

$$P(k \text{ from group } 1) = \frac{C(w,k) \times C(m,\, n-k)}{C(N,n)}$$

In plain words: count the ways to choose k people from the first subgroup, count the ways to choose the remaining (n−k) people from the second subgroup, multiply those two counts together (since any choice from the first group can be paired with any choice from the second), then divide by the total number of ways to pick any n people at all from the full population. Here, w and m are the sizes of the two subgroups (e.g., women and men); N = w+m is the total population; n is the sample size; k is how many are drawn from the first subgroup.

SAMPLE DRAWN ENTIRELY FROM ONE SUBGROUP

$$P(\text{entirely from group } g) = \frac{C(g,n)}{C(N,n)}$$

This is simply the mixed-sample formula from above with k set equal to n — every single person in the sample comes from the one group, and none from anywhere else.

A SPECIFIC INDIVIDUAL BEING INCLUDED IN THE SAMPLE

$$P(\text{specific item included}) = \frac{n}{N}$$

This last result is a helpful shortcut: it means you never need to compute a full combination just to find the chance that one particular, named person or item ends up in the sample — it is simply the sample size divided by the population size.

How to solve this type of question

1. Identify the total population size N and the sample size n being drawn.
2. Identify the size of each relevant subgroup (e.g., how many women, how many men).
3. Compute the numerator: the number of ways to satisfy the exact condition in the statement (e.g., exactly 3 from one subgroup and 2 from the other), using the combination formula for each subgroup and multiplying them together.
4. Compute the denominator: the total number of ways to choose any sample of size n from the full population N, namely C(N, n).
5. Divide numerator by denominator.

EXAMPLE 7.1 — worked calculation

A committee of 5 is selected at random from 15 people (6 women, 9 men). Find P(exactly 3 women and 2 men) and P(the committee is entirely women).

C(6,3) × C(9,2) = 20 × 36	**720**
C(15,5)	**3003**
P(3 women, 2 men) = 720 / 3003	**23.976%**
P(all women) = C(6,5) / C(15,5) = 6 / 3003	**0.1998%**
Ratio of the two probabilities: 720 / 6	**120 times**

A useful shortcut for ratios

Whenever two probabilities are computed over the same population and sample size, they share the same denominator, C(N,n). This means their ratio can be found by simply dividing the two numerators — there is no need to actually compute the shared denominator at all if only a ratio, rather than an absolute probability, is being asked for.

08 The Law of Total Probability and Bayes' Theorem

How to recognize this type of question

Several distinct sources, machines, suppliers, or groups each contribute a share of a total and each have their own rate of producing some event (e.g., defect rates, error rates). The question then reverses direction and asks: "given that the event happened, what is the probability it came from a particular source?" Any "given the outcome, find the cause" phrasing is the clearest signal for this section.

Some events can happen through more than one distinct path or source — a product might come from one of several different factories; a late delivery might come from one of several different suppliers. Each source has its own share of the total (called its "prior" probability — the probability of that source before we know anything else), and its own rate of producing the event we actually care about. The law of total probability combines all of these separate sources into one single overall rate. It is nothing more complicated than a weighted average: each source's own event rate is weighted by how big a share of the total that source represents.

Once the overall event has actually been observed — for instance, once we know a widget really is defective — Bayes' theorem lets us work backward and ask: given that this happened, how likely is it that it came from one particular source? This flips the direction of the original information around: we started out knowing "given the source, what is the event rate," and we end up finding "given the event, what is the likely source."

A probability tree, shown in Figure 8.1, is simply a diagram that makes this whole process visual and easy to follow. It starts at a single root on the left. From the root, one branch is drawn for each possible source, labeled with that source's prior probability. Then, from the end of each of those branches, two further branches are drawn — one for "the event happens" and one for "the event does not happen" — each labeled with the probability of that outcome, given the source it followed from. To find the probability of any complete path through the tree (one specific source, followed by one specific outcome), simply multiply the probabilities written along that path.

LAW OF TOTAL PROBABILITY

$$P(E) = \sum_i P(S_i) \times P(E \mid S_i)$$

In plain words: to get the overall probability of the event, go through every source one at a time, multiply that source's share of the total by its own event rate, and add up all of these products. This is exactly the same as adding up every "event happens" path in the tree.

BAYES' THEOREM

$$P(S_i \mid E) = \frac{P(S_i) \times P(E \mid S_i)}{P(E)}$$

In plain words: to find the probability that the event came from one particular source, take that one source's own path probability (prior × event rate) and divide it by the overall probability of the event (the total from the law of total probability above). S_i *is one particular source out of several possible sources; E is the observed event.*

Probability Tree — Law of Total Probability & Bayes' Theorem (Example 8.1)

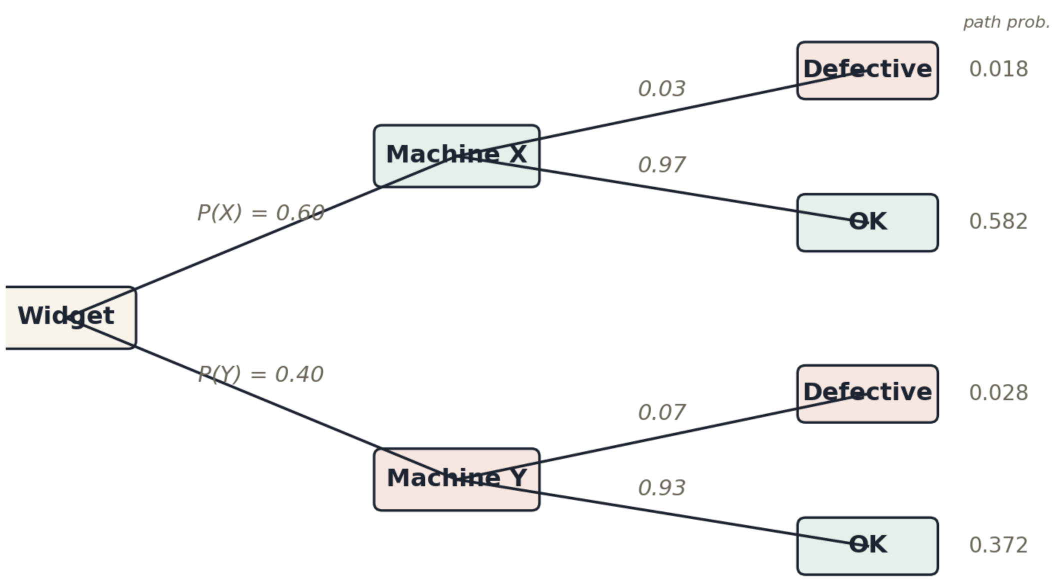

Figure 8.1 — A probability tree for Example 8.1. Multiplying along a path gives that path's joint probability (shown on the right); adding all four joint probabilities gives the overall P(Defective).

How to solve this type of question

1. Draw or list every branch of the tree: each source's prior probability, and, for each source, the probability of the event occurring given that source.

2. Multiply along each complete path (source, then event) to get that path's joint probability — these are the Bayes' theorem numerators.
3. Add together every joint probability that ends in the event actually occurring. This sum is P(Event), the law-of-total-probability result, and it is also the denominator for every Bayes' calculation that follows.
4. To find the posterior probability that the event came from a specific source, divide that source's own joint probability (from step 2) by the total from step 3.
5. If comparing two posteriors from the same event, remember they share the same denominator — so their ratio is just the ratio of the two joint (numerator) probabilities, no division by the total required.

EXAMPLE 8.1 — worked calculation

Machine X produces 60% of output with a 3% defect rate; Machine Y produces 40% with a 7% defect rate (see Figure 8.1). Find P(a widget is defective) and P(it came from Machine Y | defective).

$P(X \cap \text{defective}) = 0.60 \times 0.03$	**0.018**
$P(Y \cap \text{defective}) = 0.40 \times 0.07$	**0.028**
$P(\text{defective}) = 0.018 + 0.028$	**0.046**
$P(Y \mid \text{defective}) = 0.028 / 0.046$	**60.87%**
$P(X \mid \text{defective}) = 1 - 0.6087$ (the two posteriors must sum to 1)	**39.13%**

Note: Machine X supplies the majority of output but has the lower defect rate, so its share of the defective pool shrinks below its share of total output.

The numerators always sum to the denominator

Compute the joint probability (prior × event rate) for every source first. These numbers always add up to P(Event) — there is no separate step needed to find the total once every joint probability is known. This also means that once you know all posteriors but one, the last one follows immediately by subtracting from 1 (when there are only two sources) or from the running total (with more than two).

09 Discrete Probability Distributions

How to recognize this type of question

You are given (or must read from a chart) a full table of values a variable can take, each with its own probability — often shown as a bar chart or a table with two rows/columns, "value" and "probability." The question then asks directly for the expected value, the mean, the variance, or the standard deviation, usually by name. If a PMF table or bar chart is provided or described, this is the section to use.

A discrete random variable is a quantity produced by a random experiment that can only take specific, separate values — a count of complaints, a number of correct answers, the outcome of a die roll (as opposed to a continuous variable like height or time, which could take any value in between). The probability mass function (PMF) is the full list of every possible value the variable can take, together with the probability of each one. A PMF can be shown as a table or as a bar chart, with the height of each bar giving the probability of that particular value.

For any valid PMF, the probabilities of every possible value must add up to exactly 1 — this is a quick and easy sanity check to run before doing any further calculation, and a fast way to catch a typo or a misread table.

VALIDITY CHECK FOR ANY PMF

$$\sum_{x} P(X = x) \; = \; 1$$

In plain words: add up every probability in the table. If the total is not 1, either a value is missing, or one of the numbers has been misread.

Expected value: the long-run average

The expected value of X, written E(X) and also called the mean, represents the long-run average result if the random experiment were repeated an enormous number of times. It is found by multiplying each possible value by its own probability, and then adding up all of those products. Every value contributes to the average in proportion to how likely it is — a value that occurs often pulls the average toward itself much more than a value that occurs rarely.

MEAN (EXPECTED VALUE)

$$E(X) \; = \; \sum_{x} x \cdot P(X = x)$$

In plain words: go through every possible value x, multiply it by the probability of that value, and add up all of the results.

Variance and standard deviation: how spread out the values are

The mean alone does not tell the whole story — two distributions can share the exact same mean while looking very different, one tightly clustered around it and the other spread widely above and below it. Variance is the standard way of measuring this spread: broadly speaking, it measures how far the values in the distribution tend to sit from the mean, squaring each distance so that being above the mean and being below the mean both count as positive spread rather than cancelling each other out.

In practice, variance is computed using expected value itself, applied twice — once to X, and once to X squared:

VARIANCE

$$Var(X) \; = \; E(X^2) - [E(X)]^2$$

In plain words: find $E(X^2)$ — the expected value of X squared — the same way E(X) was found, but using x^2 in place of x in every product. Then subtract the square of the ordinary mean, $[E(X)]^2$. $E(X^2)$ means "average the squared values"; $[E(X)]^2$ means "square the average" — these are two different numbers, computed in two different orders, and must not be confused with one another.

Because variance is built from squared distances, its units are "squared" versions of whatever X measures (for example, squared complaints), which is not always easy to interpret directly. Taking the square root converts it back into the original units, giving the standard deviation — the measure of spread most often quoted and compared directly against the mean.

STANDARD DEVIATION

$$SD(X) = \sqrt{Var(X)}$$

Standard deviation is simply the square root of variance, which brings the units back in line with X itself, rather than X squared.

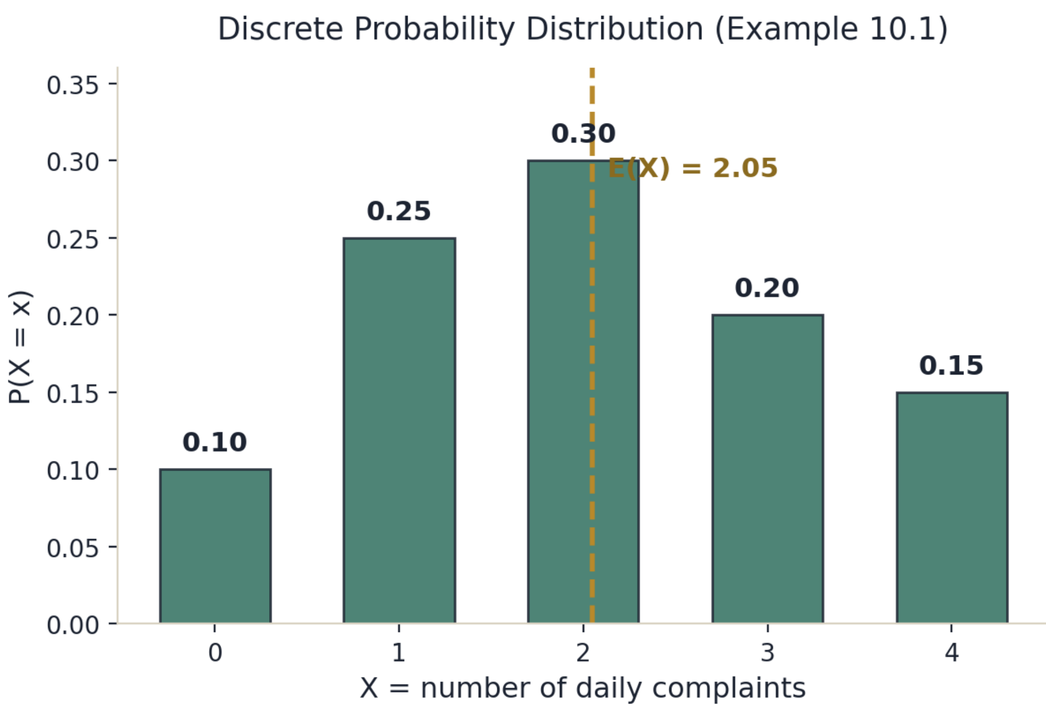

Figure 9.1 — The probability mass function for Example 9.1, with the mean E(X) = 2.05 marked.

How to solve this type of question

1. Check that every listed probability is nonnegative and that they all sum to 1 — this confirms you are reading a valid, complete PMF.
2. To find E(X): multiply each value x by its probability P(X=x), and add up all the products.

3. To find the variance: compute $E(X^2)$ the same way E(X) was found, using x^2 in place of x in every product, then subtract $[E(X)]^2$ — the square of the mean you already found in the previous step.
4. Take the square root of the variance if the standard deviation is needed instead.
5. For a cumulative probability such as $P(X \le k)$ or $P(X \ge k)$, add up the individual bar heights over the relevant range — do not assume a chart shows cumulative values directly unless it is explicitly labeled that way.

EXAMPLE 9.1 — worked calculation

A discrete variable X (the number of daily complaints at a call center) has the distribution shown in Figure 9.1: P(X=0)=0.10, P(X=1)=0.25, P(X=2)=0.30, P(X=3)=0.20, P(X=4)=0.15. Find E(X), Var(X), and SD(X).

Check: 0.10+0.25+0.30+0.20+0.15	**1.00 ✓**
$E(X) = 0(.10)+1(.25)+2(.30)+3(.20)+4(.15)$	**2.05**
$E(X^2) = 0^2(.10)+1^2(.25)+2^2(.30)+3^2(.20)+4^2(.15) = 0+.25+1.20+1.80+2.40$	**5.65**
$Var(X) = E(X^2) - [E(X)]^2 = 5.65 - 2.05^2 = 5.65 - 4.2025$	**1.4475**
$SD(X) = \sqrt{1.4475}$	**1.203**

Result: E(X) = 2.05, Var(X) = 1.4475, SD(X) ≈ 1.203

$E(X^2)$ is not the same as $[E(X)]^2$

This is the single most common variance error. $E(X^2)$ means "average the squared values" — square each x first, then weight by probability and sum. $[E(X)]^2$ means "square the average" — find the ordinary mean first, then square that one final number. Compute them in the correct order and keep them in separate columns of working, since swapping them will silently produce the wrong variance.

A bar chart shows P(X=x), not P(X≤x)

When reading a PMF bar chart, each bar's height is the probability of that exact value, not the cumulative probability up to that point. To answer a question about $P(X \le k)$ or $P(X \ge k)$, you must add together the appropriate individual bars yourself.

10 Common Errors Across This Topic

The following errors account for the majority of incorrect answers across the subtopics covered in this guide. Reviewing them alongside the worked examples above is one of the most efficient ways to prepare.

- **Reversing a conditional probability.** P(A | B) and P(B | A) answer two different questions and are generally not equal to one another. In Example 4.1, P(same number | odd product) was found to be exactly 1/3 — but P(odd product | same number) would need to be computed completely separately, starting from a different conditioning event, and would not automatically come out to the same value.
- **Treating independence and mutual exclusivity as related ideas.** Independence means one event gives no information about another. Mutual exclusivity means the two events cannot both happen. These are almost opposite conditions: two mutually exclusive events with nonzero probability can never be independent, since learning that one occurred tells you with certainty that the other did not.
- **Forgetting that "without replacement" shrinks the pool.** Each successive draw without replacement reduces both the numerator and the denominator of the relevant fraction. Treating such draws as independent, and simply reusing the first draw's probability for the second, is one of the most frequent sources of error in this topic.

- **Confusing "exactly one" with "at least one."** The union formula from Section 3 gives P(at least one of several events); it is a different, larger quantity than P(exactly one), which requires the "only-X" construction. Read exam wording carefully — "at least," "exactly," and "at most" each require a different calculation.
- **Assuming a sample-at-once scenario behaves like independent trials.** Combinatorial (hypergeometric-style) sampling in Section 7 is still, conceptually, sampling without replacement — choosing a committee of 5 from a group is equivalent to drawing 5 people one at a time without putting anyone back. Do not apply an independent-trials formula (such as a simple product of the same fraction repeated) to this kind of question.
- **Rounding before the final comparison.** Many exam statements are constructed so that the correct answer is decided in the third or fourth decimal place, or by a ratio that sits just above or below a stated threshold. Carry every intermediate quantity to full precision, and only round at the very last step, immediately before comparing to the threshold in the statement.
- **Forgetting that $E(X^2)$ is not the same as $[E(X)]^2$.** When computing variance with the shortcut formula, $E(X^2)$ means "average the squared values," while $[E(X)]^2$ means "square the average." These are two different numbers, and mixing them up is one of the most common variance errors. Squaring the mean first and then subtracting it from the sum of the plain $x \cdot P(x)$ values (rather than the $x^2 \cdot P(x)$ values) will silently produce the wrong variance.

11 Summary Reference

TASK	METHOD
Equally likely outcomes	P(E) = n(E) / n(S)
"At least one"	Use the complement: 1 – P(none)
Union of events	P(A∪B) = P(A)+P(B)−P(A∩B); extend with pairwise/triple terms for three events
"Exactly one" / "exactly two" (three events)	Build the only-X and pairwise-only regions before dividing by the population size
Given information ("given that...")	P(A|B) = P(A∩B)/P(B)
Checking independence	Compare P(A∩B) to P(A)×P(B); one mismatch disproves it
Sequential draws, no replacement	Multiply shrinking fractions: (count/total)×(count−1)/(total−1)
Sample of several items at once	Combinatorial: [C(w,k)×C(m,n−k)] / C(N,n)
Multiple sources, then reverse ("given the outcome, which source?")	Law of total probability, then Bayes' theorem
Discrete distribution summary	$E(X)=\Sigma x \cdot P(x)$; $Var(X)=E(X^2)-[E(X)]^2$; $SD=\sqrt{Var}$

12 Formula Cheat Sheet

Every formula from this guide, collected in one place, grouped by section, for quick reference during practice or review.

01 — SAMPLE SPACES AND CLASSICAL PROBABILITY

Classical probability

$$P(E) = \frac{n(E)}{n(S)}$$

02 — THE COMPLEMENT RULE

Complement rule

$$P(A^c) = 1 - P(A)$$

"At least one"

$$P(\text{at least one}) = 1 - P(\text{none})$$

03 — UNIONS AND INCLUSION-EXCLUSION

Union of two events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Only X (exactly one)

$$\text{Only X} = |X| - |X \cap Y| - |X \cap Z| + |X \cap Y \cap Z|$$

Pairwise-only (exactly two)

$$\text{Pairwise only}(X, Y) = |X \cap Y| - |X \cap Y \cap Z|$$

Union of three events (shown larger due to its length):

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

04 — CONDITIONAL PROBABILITY

Conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Equally likely case

$$P(A \mid B) = \frac{|A \cap B|}{|B|}$$

05 — INDEPENDENT EVENTS

Independence condition

$$P(A \cap B) = P(A) \times P(B)$$

Equivalent condition

$$P(A \mid B) = P(A)$$

06 — PROBABILITY WITHOUT REPLACEMENT

Two draws, same category

$$P(\text{both from X}) = \frac{\text{count}}{\text{total}} \times \frac{\text{count} - 1}{\text{total} - 1}$$

One of each, either order

$$P(\text{one X, one Y}) = 2 \times \frac{X}{\text{total}} \times \frac{Y}{\text{total} - 1}$$

07 — COMBINATORIAL PROBABILITY

Permutations (order matters)	$P(n, r) = \dfrac{n!}{(n-r)!}$
Combinations ("n choose r")	$C(n, r) = \dfrac{n!}{(n-r)!\; r!}$
Mixed sample, two subgroups	$P(k \text{ from group } 1) = \dfrac{C(w, k) \times C(m,\, n-k)}{C(N, n)}$
Entirely from one subgroup	$P(\text{entirely from group } g) = \dfrac{C(g, n)}{C(N, n)}$
One specific item included	$P(\text{specific item included}) = \dfrac{n}{N}$

08 — LAW OF TOTAL PROBABILITY AND BAYES' THEOREM

Law of total probability	$P(E) = \sum_i P(S_i) \times P(E \mid S_i)$
Bayes' theorem	$P(S_i \mid E) = \dfrac{P(S_i) \times P(E \mid S_i)}{P(E)}$

09 — DISCRETE PROBABILITY DISTRIBUTIONS

Validity check	$\sum_x P(X = x) = 1$
Mean (expected value)	$E(X) = \sum_x x \cdot P(X = x)$
Variance	$Var(X) = E(X^2) - [E(X)]^2$
Standard deviation	$SD(X) = \sqrt{Var(X)}$